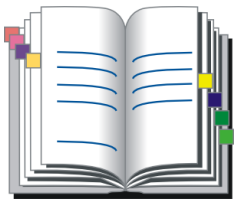
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# Faculty of Science and Engineering

## Department of Electrical and Electronic Engineering

## Project Activity Log Sheets



## ACADEMIC YEAR 20__ / 20__

## Course Code: EEEE1027 / EEEE1031

**Student's Name** : ……Tan Kai Le……

**Student's ID** : ……20675369……...

**Group Number** : ……E34……

**Course** : Please tick (√)

*Electrical and Electronic Engineering* ( √ )

*Mechatronic Engineering* ( )

**Student's Signature** :

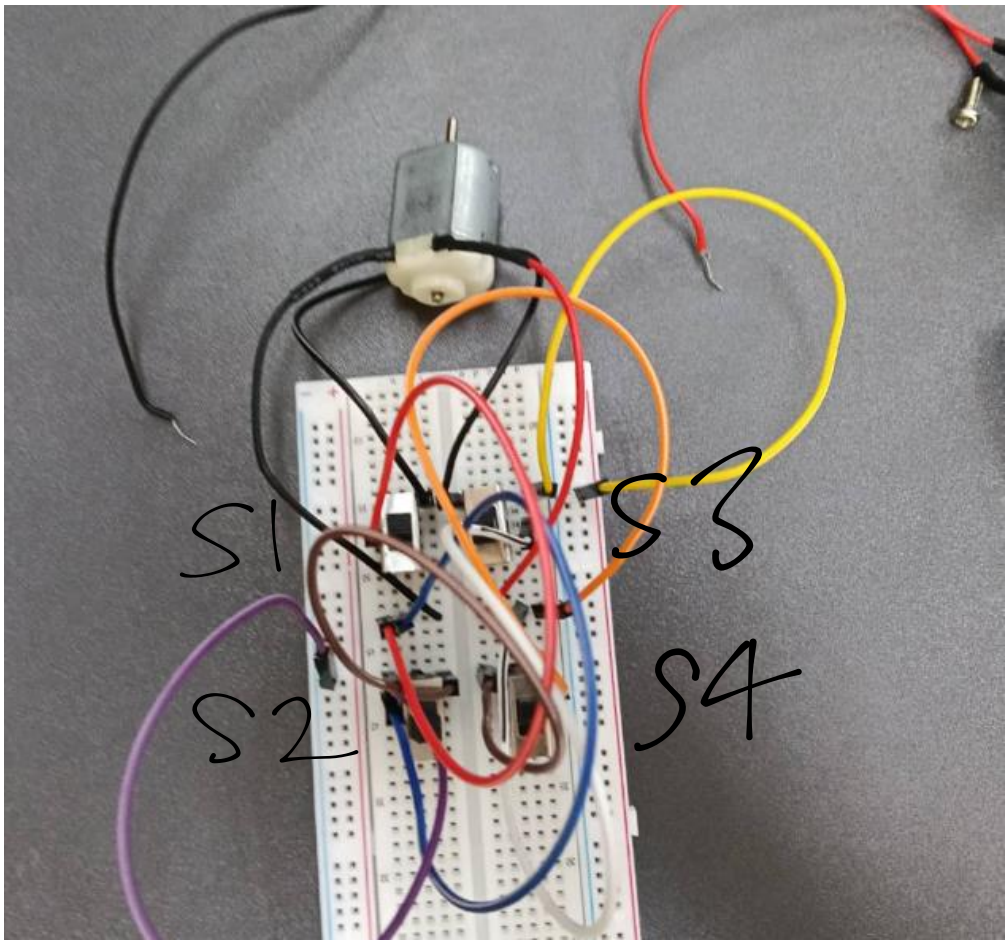
# Project Activity Log

Date (and time): 3/11/2025 4:30pm

Title of work:  H-bridge

Purpose: To turn a motor clockwise and anticlockwise using an H-bridge.

List of Equipment used: wires, DC motor, 4 switches, breadboard.




Procedure:

1. A multimeter was used to check if switches were on or off.
2. The motor's direction was controlled by turning on specific switches (S1 and S4 for clockwise, S2 and S3 for anticlockwise).
3. Applied friction to the motor and observed that the current increased.

Table 1 The H-Bridge Controller: Measuring $I_{ld}$ at no-load

Note: Replace 1.5 V by 3 V and 3 V by 6 V of column 1 in the following table if the operating voltage range that your motor is 3 – 6 V. If you are unsure, please check with lab demonstrator.

| $V_{cc}$ (V) | S1 | S2 | S3 | S4 | Direction of rotation | $I_{ld}$ (A) |
|---|---|---|---|---|---|---|
| 1.5 | 0 | 0 | 0 | 0 | — | 0 |
| 1.5 | 1 | 0 | 0 | 1 | clockwise | 0.16 |
| 1.5 | 0 | 1 | 1 | 0 | anticlockwise | −0.16 |
| 3 | 0 | 0 | 0 | 0 | — | 0 |
| 3 | 1 | 0 | 0 | 1 | clockwise | 0.25 |
| 3 | 0 | 1 | 1 | 0 | anticlockwise | −0.25 |

Table 2 The H-Bridge Controller: Measuring $I_{ld}$ with some loading

Note: Replace 1.5 V by 3 V and 3 V by 6 V of column 1 in the following table if the operating voltage range that your motor is 3 – 6 V. If you are unsure, please check with lab demonstrator.

| $V_{cc}$ (V) | S1 | S2 | S3 | S4 | Direction of rotation | $I_{ld}$ (A) |
|---|---|---|---|---|---|---|
| 1.5 | 0 | 0 | 0 | 0 | — | 0 |
| 1.5 | 1 | 0 | 0 | 1 | clockwise | 0.25 |
| 1.5 | 0 | 1 | 1 | 0 | anticlockwise | −0.25 |
| 3 | 0 | 0 | 0 | 0 | — | 0 |
| 3 | 1 | 0 | 0 | 1 | clockwise | 0.45 |
| 3 | 0 | 1 | 1 | 0 | anticlockwise | −0.45 |

11) Did the current change when friction is applied to the shaft?

Answer = Yes.

12) Now, put the switches in the required configuration to spin the motor in a clock-wise fashion. What is the current when you hold the motor shaft so it can't spin (this is called the locked-rotor test)? Do the same for when the motor tries to spin anti-clockwise.

a. $I_{ld_stall_c}$ = 0.5A.

b. $I_{ld_stall_ac}$ = −0.5A.

Results and Discussion:

Observed that voltage is directly proportional to current, where voltage increases when current increases. When the motor is rotated anticlockwise, the current values were negative. Additionally, the direction of rotation was clockwise when only S1 and S4 were

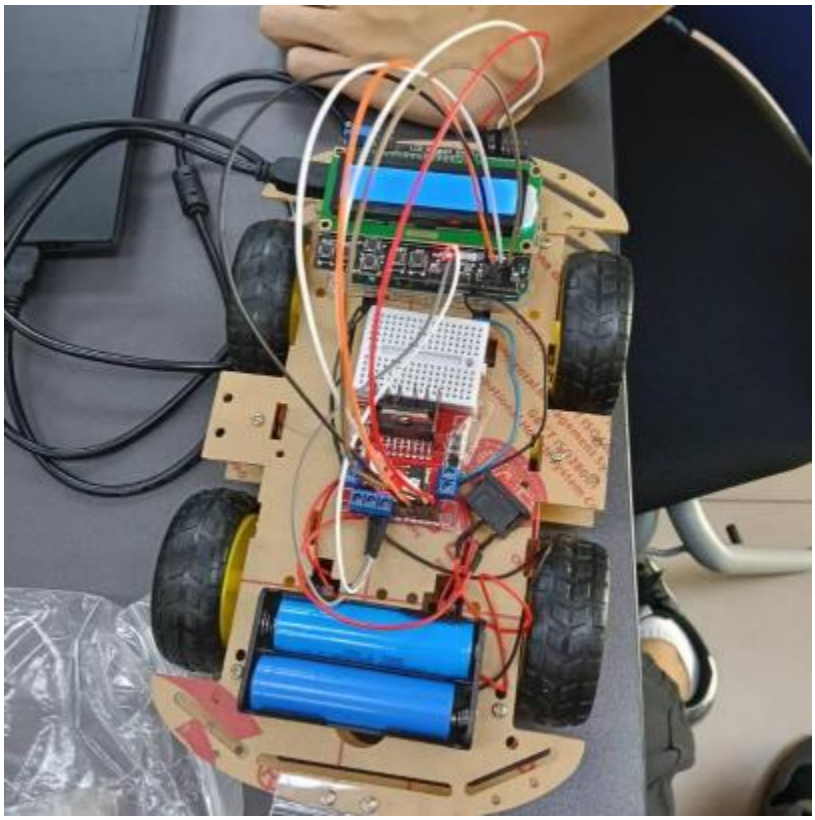
switched on, and the direction of rotation was anticlockwise when only S2 and S3 were switched on. No current and rotation was noticed when all of the switches were open. Current increased when friction was applied to the motor. Moreover, the current for maximum friction applied to the motor is 0.5A clockwise and –0.5A anti-clockwise.

Conclusion: The motor was able to turn clockwise and anticlockwise with an H-bridge with different combinations of turned on and off switches. The values of current for the corresponding combinations were successfully recorded including when maximum friction was applied to the motor.

Date and time: 10/11/2025 5:03pm

Title of work: Robot car moving in a straight line.

Purpose: To make sure that the robot car goes in a straight line for 10 seconds with an LCD display from 1 to 10 seconds.



List of equipment used: Arduino UNO with LCD keypad shield, 4 DC motors, L298N motor driver module, 2 Batteries, Robot car kit.

Procedure:

1. Assembled Components

• Mounted Arduino UNO with LCD Keypad Shield on the chassis.

- Attached 4 DC motors (2 left, 2 right) using screws and nuts.
- Installed L298N motor driver module.

2.Connected Power

- Battery + → both L298N and +12 V
- Battery – → both L298N GND and Arduino GND
- L298N 5 V OUT → Arduino 5 V

3.Wired Motor Drivers

- Driver A (Left Motors):
  - ENA → D5  IN1 → D10  IN2 → D11
- Driver B (Right Motors):
  - ENB → D6  IN3 → D12  IN4 → D13
- OUT1/OUT2 → Left motors OUT3/OUT4 → Right motors

4.Upload Code

- Used Arduino to code that:
  - drives both sides forward using PWM (analogWrite on D5, D6)
  - counted 1–10 s on LCD
  - Stopped after 10 s.

5.Run Test

- Powered the car.
- LCD showed “Moving Forward” and counts 1→10.
- Car moved straight for 10 seconds, then stopped displaying “Stopped at 10s.”
- Edited motor speed value in code (range 0–255) to tune forward speed.

Results and Discussion:

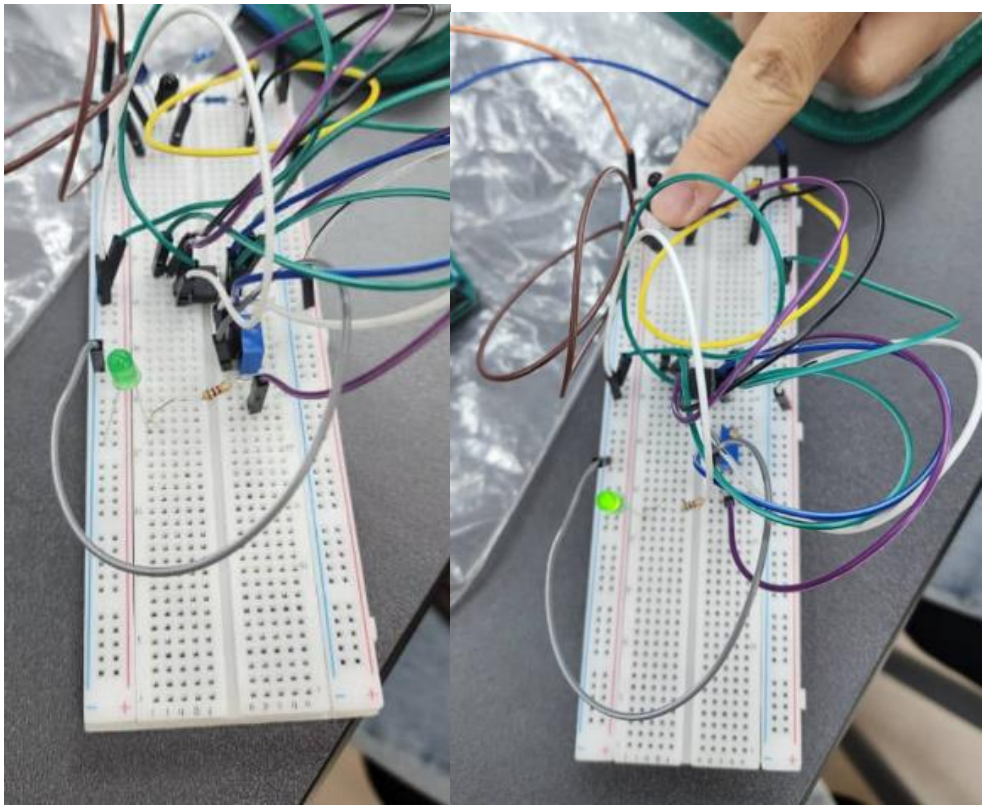
An Arduino is used to control the timing and movement of the robot car. The car moved ahead in a straight path when the program began since all four DC motors are turned on at once. The motors ran for precisely ten seconds before being automatically stopped by the Arduino, which maintained track of time. The running time, which counted from 1 to 10 seconds to represent the car's progress, is simultaneously displayed on the LCD keypad shield. When ten seconds had passed, the car stayed motionless, and the display stopped updating, signifying that the task was finished. Made sure that the motors were installed properly so that the wheels would not wobble as the car runs. Additionally, take into consideration the weight distribution of the robot car regarding where you mount the

electrical components on the robot car as it could affect the balance of the car. The PWM of the left tires of the car was set to be slower than the right side resulting in the car moving straight for at least 8 seconds.

Conclusion:
The robot car successfully moved forward in a straight line for about 8 seconds and steered off to the right. Moreover, an LCD display of the timer from 0 to 10 seconds was shown.

Date and time: 20/11/2025 5:11pm
Title of work: Prototyping a Light Sensor Circuit

Purpose: To build a light detection circuit that indicates light or dark surfaces.



List of equipment used: wires, 100k Ohm variable resistor, 390ohm resistor, 1k ohm resistor, 1M ohm resistor, green LED, photodiode, IR sensor, LM358 Single Supply Dual Operational Amplifier.
Procedure:
1. Build the IR Emitter
   • Connected the anode of the IR LED to +5 V through a 390 Ω resistor.
   • Connected the cathode of the IR LED to GND.
   • This formed the IR light source that reflects a surface.

2. Build the Photodiode Sensor

- Connected Anode → GND.
- Connected Cathode → inverting input (–) of the first LM358 op-amp.
- Placed a 1 MΩ resistor between the inverting input (–) and the output of that opamp.
- Connected the non-inverting input (+) to GND.
- Powered the LM358: Pin 8 → +5 V, Pin 4 → GND.

3. Add the Comparator (second op-amp in LM358)
   - Took the analog output from the first op-amp and connected it to the non-inverting input (+) of the second op-amp.
   - Connected a 100 kΩ variable resistor between +5 V and GND, and connected its middle pin to the inverting input (–) of the second op-amp.

4. Add LED Indicator
   - Connected output of comparator → 1 kΩ resistor → green LED → GND. • LED would turn on when reflected light is strong.

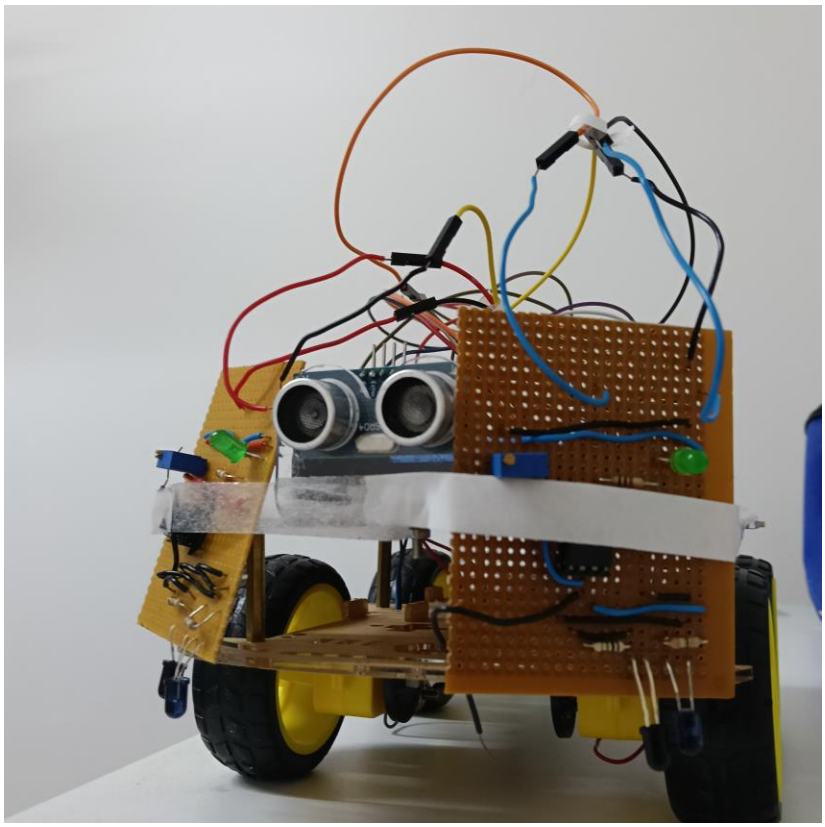
5. Testing: Blocked the Photodiode Sensor with finger to turn on the green LED.

Results and Discussion:
When the IR LED emitted light onto a surface, the amount of reflection depended on the color of the surface. A bright surface reflected more light, while a dark surface reflected very little. The photodiode detected this reflected light, and the op-amp helped to increase the signal strength so that the difference could be seen clearly. The 100k variable resistor was adjusted so that the difference between the brightness of the green LED is obvious when it is on or off. This also improved the sensitivity of the photodiode sensor.

Conclusion:
The green LED was not lit up when nothing was blocking the IR sensor, whereas the green LED lit up when blocking the IR sensor with finger. The IR emitter circuit was successfully constructed.

Date and time: 24/11/2025 4:35pm
Title of work: Line following robot car
Purpose:
1. Follow the line.
2. Complete the race in 60 seconds.

3. Display distance on LCD.



List of equipment: 2 Rotary Encoders, 2 light sensor veroboard circuits and the robot car constructed in project week one.

Procedure:

1.Design and solder two light sensor Veroboard circuits with op amp, IR, LED, photodiode, op-amp, 3 different valued resistors and a variable 100k resistor.
2.Connect them to the Arduino using the breadboard for +5V and GND.
3.Code that when the light sensor detects black, the wheels on the side of the corresponding IR sensor stops moving forward while the other side keeps moving.
4.Code that when both IR sensors detect black, both sides stop moving.
5.Connect the encoders to the arduino.

Results and Discussion:

The robot car managed to complete the track within 45 seconds and displayed the correct length of the track which is 6 metres. Observed that the IR sensors could not detect the black line because the forward speed is too fast, thus my teammate reduced the PWM value on all motors. It was successfully turning while moving along the curved line but

failed to turn fully 90 degrees on the track, to solve this problem we increased the PWM value on the turning motors.

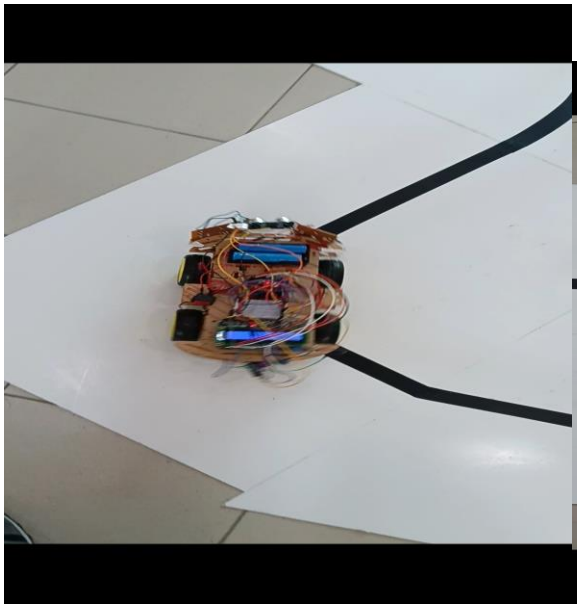
Conclusion:

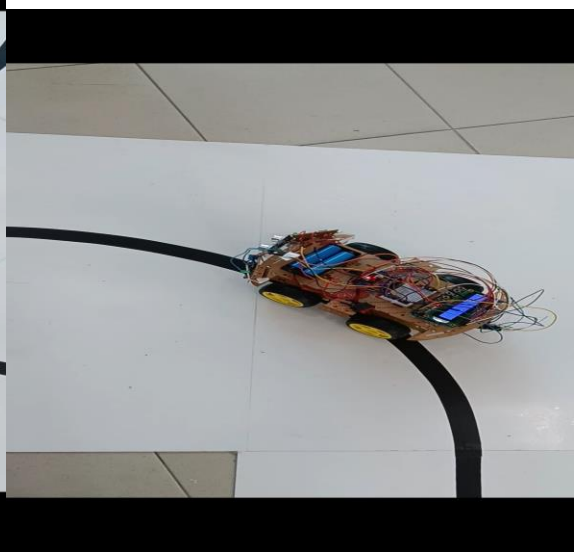
The robot car successfully completed the track within the 60 seconds time limit and a LCD display of time and distance.

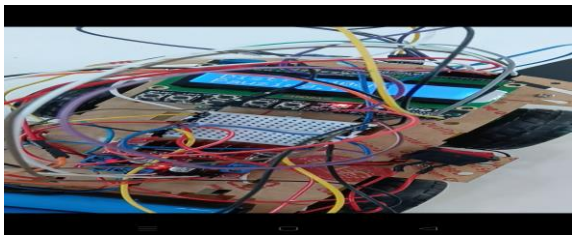
Date and time: 2/12/2025 2:35pm

Title of work: Line following with timed stop & LCD display

Purpose: To build a line following robot car that follows the path for 340 cm (according to group number), then stops for 3 seconds which should be displayed on LCD.







List of equipment: robot car constructed in project week 2

Procedure:

1. Update code from the line following task in project week 2 that the robot must stop after travelling for 340cm and stops for 3 seconds.
2. Code that the robot proceeds to finish the track after stopping at 340cm.

Results and Discussion:

The robot car successfully followed the line until the distance of 340cm and stopped for 3 seconds then continued following the line to the end of the track. Observed that there was

a delay when stopping at the desired distance by 5 cm so we coded that the car would start to stop at 335cm.

Conclusion:

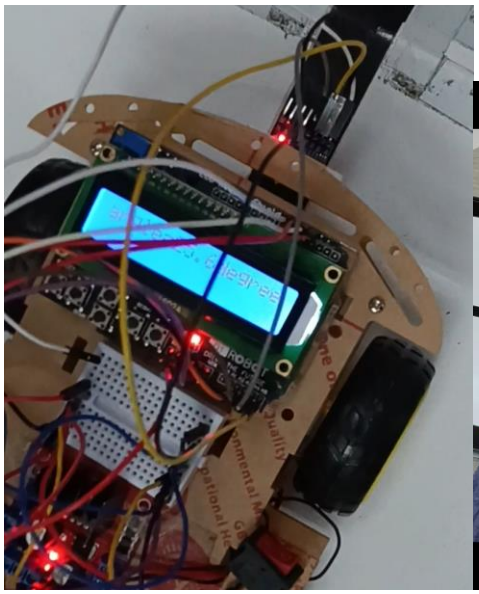
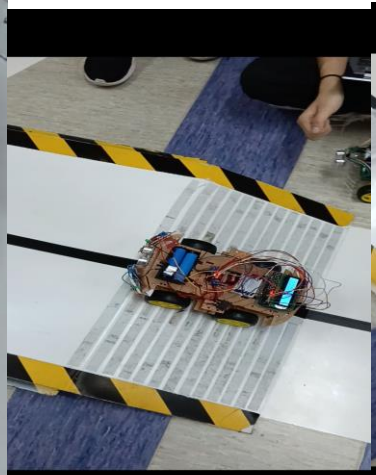
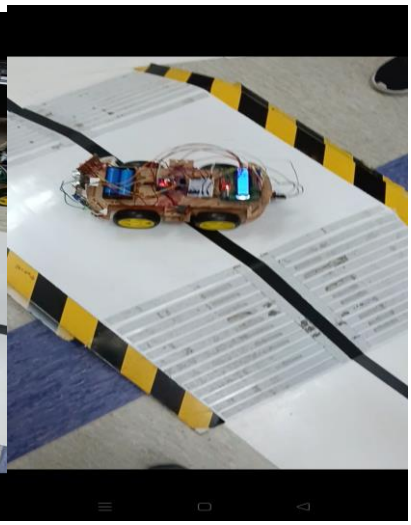
The robot car successfully followed the line until the distance of 340cm and stopped for 3 seconds then continued following the line to the end of the track.

Date and time:2/12/2025 3:39pm

Title of work: Orientation Sensing with Ramp Navigation & LCD Display

Purpose:

1. Follow the path with a ramp.
2. Display the ramp angle.
3. Stop at the top of the ramp for 4 seconds and rotates for 360˚.

List of equipment: robot car built in project week 2, MPU-6050 Gyro Accelerometer sensor.

Procedure:

1. code that the robot car moves forward with 200 pwm.

2.Connect the MPU-6050 to the Arduino

- VCC to 5V
- GND to GND
- SDA to A4
- SCL to A5

- INT to pin 2

3. Apply trigonometry to the code so that it can measure angle of ramp.

4. code that after the MPU detects the angle goes back to 0 degrees, it stops for 4 seconds and an LCD display of the angle of the ramp shown.

5. code that it rotates 360 degrees and moves forward down the ramp for 2 seconds.

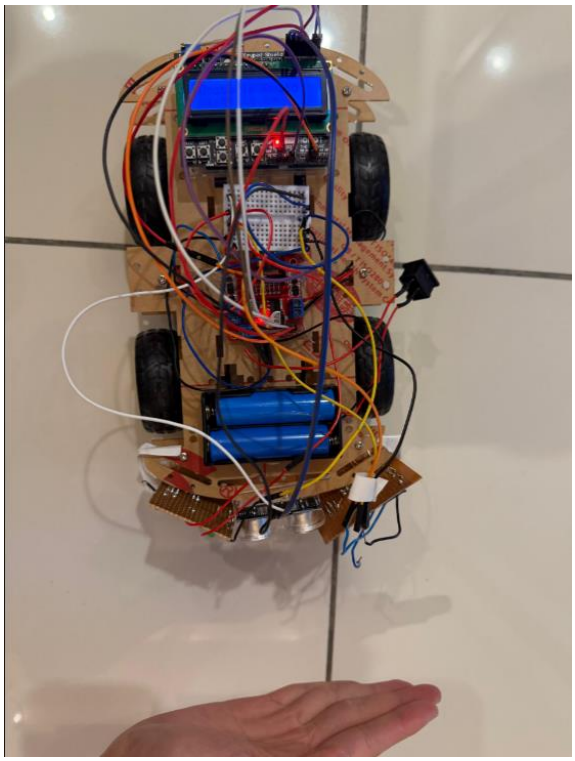
Results and Discussion:

The robot car was able to move up the ramp and an LCD display of the angle of the ramp (25.6 degrees) was shown as it stopped for 4 seconds. Moreover, it successfully rotated 360 degrees before moving down the ramp. Observed that the robot car moved past the top of the ramp and could not detect the accurate angle of the ramp when using the maximum PWM value of 255. Thus, we decided to decrease the PWM value to 200. There were two ramps that we could use for the task and the ramp with strips of rubber tape attached to it increased the friction of the ramp and assisted the robot car to move up the ramp with ease.

Conclusion: The robot car was able to move up the ramp and an LCD display of the angle of the ramp (25.6 degrees) was shown as it stopped for 4 seconds. Moreover, it successfully rotated 360 degrees before moving down the ramp.

Date and time: 3/12/2025 12:25pm

Title of work: Obstacle Detection with Autonomous Response



Purpose: To make robot car reroute after detecting any obstacles.

List of equipment: Robot car constructed in project week 2, SN-HC-SR04 ultrasonic sensor.

Procedure:

1.Connect Ultrasonic sensor to Arduino

- VCC to 5V
- GND to GND
- Trig and Echo to any pin

2.code that after detecting an obstacle

- it reverses
- turns left until no obstacle detected
- Continues to move forward

Results and Discussion:

The robot car successfully detected the obstacle and rerouted to avoid the obstacle. When a reflected pulse is detected, the Echo pin goes down. The length of time when the Echo pin was HIGH may be used to calculate the distance between the sensor and the item. The formula used to calculate the distance between ultrasonic sensor and obstacle was

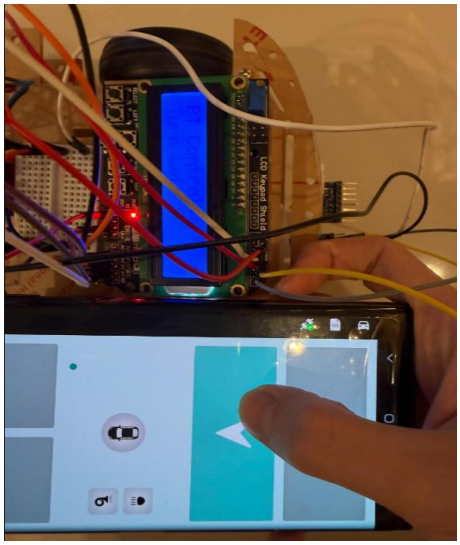
$$Distance = \frac{(Speed \times Time)}{2} = \frac{(0.034\ cm/\mu s \times Time)}{2}$$

Conclusion:

The robot car successfully detected the obstacle and rerouted to avoid the obstacle.

Date and time: 3/12/2025 3:36pm

Title of work: Bluetooth-Controlled Behaviour



Purpose: To control robot car using Bluetooth remote controller app on phone.

List of equipment: Robot car construction in project week 2, HC-05 Bluetooth Serial Module.

Procedure:

1.Connect HC-05 to Arduino

- VCC to 5V
- GND to GND
- TXD to pin 10
- RXD to pin 11

2. Code that the robot car follows the commands of the Bluetooth remote controller app

- Move forward
- Move backward
- Turns right
- Turns left

Results and Discussion:

The Bluetooth remote controller app was able to command the robot car to move forward, backward, turn left and right. Observed that Arduino TX outputs 5V, but HC-05 RX expects 3.3V so we used a voltage divider to protect the RX pin. To decrease from Arduino TX (5V) to HC-05 RX (≈3.3V), this formula was used:

$$V_{out} = V_{in}\left(\frac{R_2}{R_1 + R_2}\right)$$

Where:

• $Vin = 5\ V$

• $Vout \approx 3.3\ V$

• $R1$connected to Arduino TX

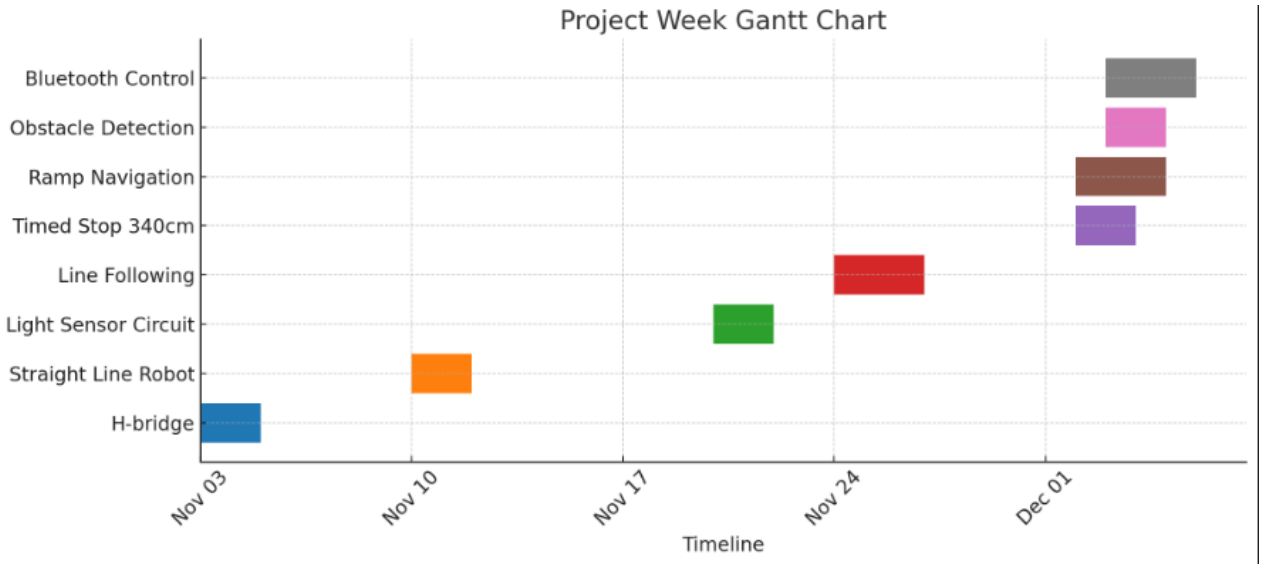
• $R2$connected to GND

• Output is taken between $R1$ and $R2$

Conclusion:

The Bluetooth remote controller app was able to command the robot car to move forward, backward, turn left and right.

Appendix:




Code List: https://github.com/kyletan06/arduino_robot_car